

ARKYA CHATTERJEE

www.mit.edu/~achatt/

EDUCATION

Massachusetts Institute of Technology (MIT) *May 2025 (expected)*
Ph.D. in Condensed Matter Theory, Department of Physics

Indian Institute of Technology Bombay (IIT-B) *May 2019*
B.Tech. (with Honors) in Engineering Physics and Minor in Mathematics

PUBLICATIONS AND PREPRINTS

Z. D. Shi, **A. Chatterjee**, *Analytic framework for self-dual criticality in $\mathbb{Z}_k$ gauge theory with matter*, [arXiv:2407.07941](#) (submitted)

A. Chatterjee, Ö. M. Aksoy, X.-G. Wen, *Gapped Phases and Phase Transitions in Spin Chains with Exact Noninvertible Symmetries*, [arXiv:2405.05331](#) (submitted)

I. T. Rosen, S. Muschinske, C. N. Barrett, **A. Chatterjee**, *et al.*, *Implementing a synthetic magnetic vector potential in a 2D superconducting qubit array*, [arXiv:2405.00873](#) (submitted)

A. Chatterjee, W. Ji, X.-G. Wen, *Emergent Generalized Symmetry and Maximal Symmetry-Topological-Order*, [arXiv:2212.14432](#) (submitted)

A. Chatterjee, X.-G. Wen, *Holographic Theory for Continuous Phase Transitions: Emergence and symmetry protection of gaplessness*, *Phys. Rev. B* **108**, 075105 (2023)

A. Chatterjee, X.-G. Wen, *Symmetry as a Shadow of Topological Order and a Derivation of Topological Holographic Principle*, *Phys. Rev. B* **107**, 155136 (2023)

M. Chatterjee, **A. Chatterjee**, A. Nandi, A. Sain, *Dynamics and stability of contractile actomyosin ring in the cell*, *Phys. Rev. Lett.* **128**, 068102 (2022)

A. Fischer, **A. Chatterjee**, T. Speck, *Aggregation and sedimentation of active Brownian particles at constant affinity*, *J. Chem. Phys.* **150**, 064910 (2019)

INVITED TALKS

Quantum Matter in Mathematics & Physics seminar *June 2024*
Harvard University Center of Mathematical Sciences and Applications

[Symmetry Seminar](#) (virtual) *May 2024*

Condensed Matter Theory (CMT) seminar, Indian Institute of Science *April 2024*

CMT *Kids'* seminar, Harvard University *February 2024*

CMT seminar, Boston University *November 2023*

Physics colloquium, IIT-B *June 2023*

CMT seminar (virtual), Pennsylvania State University *April 2023*

Category and Topological Order seminar (virtual) *November 2022*
Beijing Institute of Mathematical Sciences and Applications

CONFERENCES AND SUMMER SCHOOLS

Paths to Quantum Field Theory 2024: short talk University of Sarajevo	<i>July 2024</i>
Prospects in Theoretical Physics 2024 summer school: poster & gong show Institute for Advanced Study	<i>July 2024</i>
Summer school on “Symmetries and Anomalies: a modern take”: poster & gong show Institut des Hautes Études Scientifiques	<i>June 2024</i>
Higher Categorical Tools for Quantum Phases of Matter: gong show Perimeter Institute for Theoretical Physics	<i>March 2024</i>
Simons Ultra-Quantum Matter collaboration meeting: poster University of Colorado Boulder	<i>May 2023</i>

SELECTED HONORS AND FELLOWSHIPS

Invitation to the KITP program on “Generalized Symmetries in Quantum Field Theory”	<i>2025</i>
C. M. Clay and H. W. Kendall Fellowship, MIT	<i>2019</i>
K. Seshia Research Excellence Award for best undergraduate thesis, IIT-B Physics	<i>2019</i>
DAAD WISE fellowship for a research internship in Germany	<i>2018</i>
Institute Academic Prize for highest GPA in the Physics department, IIT-B	<i>2018</i>

TEACHING AND MENTORING

Massachusetts Institute of Technology, Cambridge, MA

- Teaching Assistant for quantum, statistical, and classical mechanics classes at undergraduate & graduate levels; conducted weekly recitations and contributed to setting and grading exams.
- Mentored undergraduate students as part of the Physics Directed Reading Program

Indian Institute of Technology Bombay, Mumbai, India

- Teaching Assistant for first-year undergraduate electromagnetism and calculus classes
- Mentor for the Summer of Science reading projects

SERVICE AND OUTREACH

Project SHORT , Mentor	<i>2024 – present</i>
Responsible for providing 1-on-1 graduate school application assistance to prospective Physics PhD candidates; aimed at shrinking the socioeconomic gap in graduate school	
SciPost Physics and Physical Review B , Referee	<i>2023 – present</i>
Refereed articles on condensed matter and quantum information physics	
MIT Physics Graduate Students Council , Webmaster	<i>2020 – 22</i>
Responsible for maintaining the PGSC website, providing web support to the Physics Values Committee	

Harvard Museum of Natural History, Volunteer *2022*
Gallery facilitator for Glass Flowers, Maya Civilization, and Sharks exhibits

IIT-B Maths & Physics Club, Invited Speaker *2020*
Gave a (virtual) pedagogical talk to an undergraduate audience on the quantum Hall effect

IIT-B Physics Academic Mentorship Program, Co-Lead *2018 – 19*
Led a team of 12 mentors to provide academic support and facilitate outreach to ~ 200 students